

PUNTA ARENAS, Chile

WMO: 859340

Lat: 53.0033S

Lon: 70.8450W

Elev: 125

StdP: 14.63

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	25.9	17.7	13.5	28.4	20.4	15.4	30.0	40.0	42.0	35.3	40.9	6.2	260	0.896

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	65.0	54.2	62.5	52.8	60.4	51.5	55.9	62.7	54.1	60.4	52.4	58.3	19.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.2	58.2	57.4	50.4	54.4	55.9	48.8	51.3	54.6	23.9	62.8	22.7	60.3	21.8	58.1	66.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 39.4	34.1	30.6	DB	16.8	72.5	3.9	4.0	14.0	75.4	11.7	77.8	9.5	80.1	6.7	83.0	(4)
(5)			WB	16.0	60.0	3.8	2.5	13.3	61.8	11.1	63.3	9.0	64.7	6.2	66.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	51.4	50.9	48.3	44.0	39.5	35.6	35.3	37.1	40.5	43.9	47.0	49.8
	DBStd	7.06	3.94	4.57	4.47	4.31	4.37	4.64	4.45	3.87	3.88	3.94	4.07	3.85
	HDD50	2585	30	38	85	186	326	431	454	401	284	192	106	52
	HDD65	7821	423	395	517	630	791	881	919	866	734	654	539	472
	CDD50	239	72	63	33	6	0	0	0	0	0	3	17	45
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	2	1	1	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	14.7	17.0	15.5	14.6	13.8	12.4	12.1	12.7	13.4	14.0	16.2	17.4	17.3
(15) Precipitation	PrecAvg	14.8	1.5	1.2	1.4	1.7	1.5	1.2	1.1	1.2	0.9	1.0	1.0	1.2
	PrecMax	23.7	3.7	3.3	3.9	3.7	3.8	3.5	3.7	3.6	2.4	3.0	2.7	4.6
	PrecMin	8.4	0.2	0.1	0.3	0.4	0.2	0.1	0.1	0.2	0.2	0.2	0.2	0.1
	PrecStd	3.1	0.8	0.7	0.8	0.9	0.8	0.7	0.8	0.7	0.6	0.6	0.6	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.4	70.2	66.2	59.4	53.5	50.0	48.4	49.9	56.8	59.4	64.2	65.9
		MCWB	55.9	58.2	54.7	52.5	48.1	46.2	43.6	43.8	47.2	48.4	51.8	53.6
	2%	DB	64.4	65.5	61.2	55.6	50.0	46.5	44.9	46.8	52.0	55.8	59.7	62.3
		MCWB	53.5	55.8	52.7	50.0	45.7	43.0	41.3	41.9	44.5	46.8	50.1	51.8
	5%	DB	61.1	61.3	58.6	53.4	47.9	44.3	43.1	44.8	49.8	53.7	57.1	59.3
		MCWB	52.0	53.0	51.0	48.3	44.4	41.7	40.1	40.8	43.4	45.5	48.3	50.1
	10%	DB	59.0	58.8	55.7	50.7	46.2	42.5	41.3	42.9	47.3	51.7	55.0	57.3
		MCWB	50.6	51.0	49.3	46.5	43.4	40.5	38.9	39.5	42.0	44.4	46.9	48.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.5	60.3	56.8	54.3	49.1	46.7	44.2	44.9	48.0	50.6	53.9	55.6
		MCDB	65.7	67.4	63.4	57.4	51.6	49.3	46.8	48.0	55.6	57.0	60.9	62.8
	2%	WB	55.0	56.6	53.8	51.1	46.4	43.3	41.9	42.6	45.4	48.2	51.3	53.2
		MCDB	62.0	63.7	60.2	54.7	48.7	45.0	43.9	45.3	50.3	54.8	57.8	59.8
	5%	WB	53.2	54.2	51.8	49.1	44.8	41.7	40.4	41.1	43.6	46.3	49.5	51.2
		MCDB	59.8	60.2	56.7	52.3	46.5	43.2	42.2	43.6	47.8	51.7	55.3	57.6
	10%	WB	51.5	52.0	50.3	47.2	43.5	40.2	39.1	39.8	42.2	44.9	47.8	49.8
		MCDB	57.5	57.4	54.5	50.0	45.2	41.8	40.9	42.2	45.9	49.9	53.6	55.8
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	14.0	13.2	11.5	10.0	8.9	9.0	9.8	11.9	13.7	14.0	14.2
		MCDBR	16.9	17.9	16.5	14.3	11.6	10.0	9.5	11.7	15.2	16.6	17.0	16.8
		MCWBR	9.8	10.8	10.3	10.2	8.8	8.1	7.6	8.3	9.6	9.8	9.8	9.7
	5% WB	MCDBR	16.1	16.7	14.7	13.3	10.1	9.3	9.0	10.7	13.6	15.4	16.0	16.0
		MCWBR	10.3	11.2	10.2	10.4	8.5	8.1	7.8	8.2	9.3	9.9	10.0	10.4
(40) Clear-Sky Solar Irradiance	taub		0.325	0.318	0.319	0.310	0.290	0.278	0.288	0.301	0.316	0.319	0.322	0.325
	taud		2.548	2.575	2.558	2.544	2.489	2.391	2.436	2.518	2.509	2.518	2.522	2.531
	Ebn at Noon		311	302	281	250	216	193	211	249	277	298	310	314
	Edh at Noon		32	29	25	20	16	14	16	20	26	30	32	33
(44) All-Sky Solar Radiation	RadAvg		1699	1397	940	539	290	185	234	445	829	1311	1643	1801
	RadStd		102	81	57	32	26	16	16	30	59	64	91	110

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+1.12	N/A	N/A	N/A	-1.35	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	+1.35	N/A	N/A	N/A	-0.72	-151	N/A	N/A	N/A

Nomenclature: See separate page